

38 Two neutral atoms are isotopes of the same element.

Which statement about the atoms is correct?

- A They have a different number of neutrons and a different number of electrons.
- B They have a different number of neutrons and the same number of protons.
- C They have the same number of neutrons and a different number of electrons.
- D They have the same number of neutrons and the same number of protons.

39 What is the composition of a meson?

- A 1 quark and 1 antiquark
- B 1 quark and 2 antiquarks
- C 2 quarks and 1 antiquark
- D 3 quarks

40 A nucleus of carbon-11 contains 6 protons and 5 neutrons.

The nucleus of carbon-11 decays by β^+ emission.

What is the total number of up and down quarks in the product of the decay of this nucleus?

	up quarks	down quarks
A	15	18
B	16	17
C	17	16
D	18	15

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